

AIMScribe v3 — UIU AIMS LAB Server Requirements

For the Curaxx server · AIMS LAB, United International University · 29 September 2026

AIMScribe records doctor–patient consultations in seven partner clinics and files each recording on this server with its clinical record beside it. Every figure below is **measured on our own bench**, not estimated: 28 rooms recording at once (deliberately twice the real load), 2,434 audio pieces, 6.1 GB in a single burst — and a full consultation carried end to end, from the microphone to the archive.

1 · The load this server must carry

Figure	Value
Clinics	7 — six Aalo Clinic branches and Amader Susastho
Consulting rooms	14
Laptops	14, one per room
Doctors	30 , sharing those 14 rooms across two shifts
Shift 1	08:30 – 14:20
Shift 2	15:00 – 21:00
Clinic days	six days a week
Consultation length	Aalo max 10 min · Amader Susastho 15–20 min
Recordings at once	14 — one per room, never 30
Study target	20,000 consultations

Size by rooms, not by doctors. A room records one consultation at a time, so 30 doctors across two shifts in 14 rooms produce 14 simultaneous recordings. Every figure here follows from the room count.

Derived, at 5.02 MB per minute (measured)	Value
Room-hours per day	166
Audio per day, rooms fully occupied	~50 GB
Audio per day, realistic occupancy	30 – 35 GB
The 20,000-consultation study	~1.15 TB on the archive
A full year at these hours	10 – 16 TB
Both databases together	0.5 MB per recording — 50 GB with indexes
Request rate, all 14 rooms	~1.2 per second, peaking near 4

This is a storage and bandwidth system, not a processor-heavy one. One request a second is trivial for any server. 30–50 GB of audio every clinic day is not, and it arrives whether anyone is watching or not.

2 · Drive addresses — the three storage areas

These are separate on purpose (SRS-SRV-03). A large merge must never slow a database write that a recorder is waiting on. Each is a mount point, set once in `deploy/uiu/.env`.

Variable	Mount address	What lives there	Size
AIMS_ARCHIVE_PATH	/srv/aims/archive	Every recording: the WAV, its manifest, its clinical JSON	1.5 TB min 2 TB comfortable
AIMS_DB_PATH	/srv/aims/db	PostgreSQL — aims_recordings and aims_clinical	on the 1 TB NVMe

AIMS_BACKUP_PATH	/srv/aims/backups	Nightly encrypted database backups	100 GB
AIMS_WORK_TMPFS	in memory (tmpfs)	One consultation being joined, compressed and encrypted	1 GB on a 4 GB box

Inside the archive, one folder per consultation:

```
/srv/aims/archive/<clinic>/<doctor>/<date>/<name>/
    <name>.wav · <name>.manifest.json · <name>.json · _index.json
```

A real example from the bench, with the clinics as they are now registered:

AALO_DH0LPUR/DR0042/2026-09-29/P0099001_DR0042_AALO_DH0LPUR_000621_000752_20260929/. The date folder is the **clinic's local date**, not UTC — an evening consultation at UTC+6 would otherwise be filed under the previous day, permanently.

AIMS_WORK_TMPFS matters more than it looks. tmpfs pages count against RAM, so on a 4 GB machine the template's 6 GB would reserve more than every container ceiling combined. Measured against 10- and 20-minute consultations, one join peaks near 320 MB, so 1 GB is already generous.

3 · The machine

Part	Minimum that works	Recommended
Cores	8	16
Memory	8 GB on Linux	32 – 64 GB ECC
System / database disk	1 TB NVMe	2 TB NVMe, mirrored, power-loss protected
Archive storage	1.5 TB usable	4 TB usable, mirrored
Internet	50 Mbit/s symmetric	100 Mbit/s symmetric
Power	UPS	UPS with automatic clean shutdown

How much archive storage, honestly

The study is **1.15 TB**, and that is the number that matters. 20,000 consultations at the agreed caps average 11.4 minutes, and audio is a measured 5.02 MB per minute. Add the 20 GB the archive worker keeps free by refusing to write below it, and the free space no volume should run without: **1.5 TB usable. Two is comfortable.**

A 12 TB array was recommended earlier. That was room to keep running for years afterwards, not what the study needs, and the two were not separated clearly enough. They are separated here:

Option	Usable	Survives a disk failure	Holds
1 × 2 TB	2 TB	No — see below	The study, with room to spare
2 × 2 TB, RAID 1	2 TB	Yes	The study, and keeps working through a failure
2 × 4 TB, RAID 1	4 TB	Yes	The study plus 2–3½ months more
4 × 6 TB, RAID 10	12 TB	Yes	The study plus about a year

Any of these can run the study. The difference is only what happens afterwards.

On running without RAID. RAID protects *availability* — it keeps the system recording through a failed disk. It is not what protects the data, and that distinction decides whether a single disk is acceptable.

Every archived recording also gets a **lossless FLAC copy in cold storage at a second provider**, encrypted here first with a key that never leaves UIU. It is not assumed to be good: the worker decodes the FLAC again and compares the samples against the archived audio before the copy counts. So a single disk is **not** a single copy — it is one of two, and the second is off-site.

If there is no RAID, two things stop being optional: the cloud copy must be **switched on and confirmed** — the dashboard shows anything not yet safe in two places, and that count should be zero every morning — and the

monthly restore test must actually happen. A copy nobody has ever read is not a backup. With those in place, a single 2 TB disk is a defensible way to run the study. Without them, neither is RAID: it has never protected anyone from a deletion or a corrupted write.

What we know about the Curaxx machine: 8 cores, 8 GB, storage to be confirmed. **Please tell us what disks it has.** Memory can be worked around and we have measured exactly how — see section 5. Storage cannot be conjured: if that box has a single 1 TB disk it can run the system but cannot hold the study, and something must be added before collection starts.

The thing that actually decides long-term cost is retention, not capacity. Even 12 TB is about a year at these hours. Once a recording has a verified off-site copy, removing the local WAV after an agreed period is safe — and that decision is worth more than any amount of disk.

Why ECC memory. The system's whole purpose is proving recordings were not altered. Uncorrected memory errors undermine that claim at the source. **Why power-loss protection on the database disk.** Consumer SSDs acknowledge a write while it is still in volatile memory and lose it in a power cut — the exact failure the integrity chain exists to detect.

4 · Operating system: Ubuntu Server 24.04 LTS — decided

AIMS LAB has chosen Linux. On an 8 GB machine it is worth about **4 GB of usable memory**: Windows plus Docker Desktop's virtual machine consume roughly half the box before AIM Scribe starts, where Ubuntu gives AIM Scribe about 7.5 GB and faster disk I/O.

During load testing on Windows, local object storage under memory pressure failed and **took the whole Docker engine down with it** — not one container. That is the strongest single argument for Linux on a small machine.

Day-to-day operation needs no Linux skill: everything is inspected through web pages from an ordinary Windows laptop. Linux knowledge is needed only to repair the machine itself, so **we need to know who at CITS supports a Linux host when the OS or hardware fails.** Required software: Docker Engine with the Compose plugin, and root/sudo on the host.

5 · What actually runs, and what it uses

Measured under the 28-room storm, not at rest:

Container	Idle	Under load	Ceiling set
The server (api)	180 MB	320 MB	1 GB, 2 workers
PostgreSQL	60 MB	240 MB	1 GB
Archive worker	22 MB	~250 MB mid-merge	768 MB
Redis	13 MB	23 MB	256 MB
Gateway (TLS)	—	~40 MB	256 MB
PgBouncer	—	~15 MB	128 MB

On a 4 GB machine, these settings are not optional:

```
AIMS_PG_CONF=. /postgres/postgresql.4gb.conf
AIMS_PG_MEMORY=1g    AIMS_PG_SHM=256m
AIMS_API_MEMORY=1g    AIMS_SERVER_WORKERS=2
AIMS_ARCHIVE_MEMORY=768m    AIMS_ARCHIVE_CPUS=2
AIMS_REDIS_MEMORY=256m    AIMS_WORK_TMPFS=1g
```

The default PostgreSQL config is sized for the recommended 64 GB machine and asks for `shared_buffers = 16GB`; on 4 GB it **will not start at all**. The small profile gives up caching and gives up nothing about durability — every `fsync` and `commit` setting is identical.

No local object storage on a small machine. Tested rather than assumed, and it failed: capped at 768 MB it could not take the storm. Pieces in transit go to Cloudflare R2 instead, which is the design anyway — R2 for pieces, cold storage for copies, the finished WAV on UIU's own disk.

6 · What the server will store

This matters for classification: **most of it is patient-identifiable health information**. A voice recording identifies a person even with no name attached.

Held	Contents	Sensitivity
<name>.wav	The consultation audio. WAV, 44.1 kHz, mono, 16-bit	PHI
<name>.json	Patient ID and name, sex, age, doctor, clinic, times, paramedic observations, diagnoses, prescription, investigations, previous visit	PHI — highest
<name>.manifest.json	Hashes, the piece list, the signed integrity chain	Internal
aims_recordings	Sessions, chains, receipts, devices, audit log. No patient names by design	Internal
aims_clinical	Patients, encounters, demographics, prescriptions, and an append-only log of who read what	PHI

Not on UIU disks: audio pieces in transit live in Cloudflare R2 and are deleted once verified here; one lossless FLAC copy goes to cold storage at a second provider, **encrypted at UIU before upload with a key that never leaves UIU**. Nothing is kept on the doctor's PC: its copy is deleted the moment this server proves it holds a good one — normally within seconds.

Retention is a decision we need from UIU, not from us. At 10–16 TB a year it is the single largest cost in the system, and it must be stated before final storage is purchased. Audit logs are kept indefinitely and are never edited.

7 · Network and firewall

Direction	Port	Purpose	Source
Inbound	443 TCP and UDP	Everything: recorders, CMED's server messages, the dashboard. UDP carries HTTP/3	Internet, or UIU network only if all clinic PCs are on it
Inbound	80 TCP	Certificate renewal only; redirects to 443	Internet
Inbound	22 TCP	Administration	Named UIU machines only
Outbound	443	Cloudflare R2, cold storage, container registries	—
Outbound	123 UDP	NTP — mandatory	—

Nothing else is published. One gateway holds the only open ports; every other service is reachable from nowhere but inside the machine. **Port 6000 must not be proposed** — browsers and Node both refuse it. The system uses 6060 internally and 443 at the edge.

Five policy points that will break the system if missed

#	Requirement	Why
1	A DNS name, never an IP, and lower case	Upload authorisation is signed with the host inside it: one capital letter refuses every upload. A changed DHCP lease already silenced every recorder once
2	No TLS interception or transparent proxy	It breaks certificate validation and can invalidate signed upload URLs
3	No aggressive idle-connection timeout	Recorders reuse a connection every 30 s and the server holds them for 75

4	Clock synchronisation (NTP)	Recordings are filed under the clinic's local date, decided once. A drifting clock misfiles consultations permanently
5	A static address, or a documented permanent reservation	Certificates and recorder configuration both depend on it

If CITS will not open an inbound port, the design tolerates it: the archive component opens no ports at all and makes only outbound connections. For the parts that must be reachable, a Cloudflare Tunnel or Tailscale works with no inbound rule whatsoever — the server dials out. We are happy to take that route if it is easier to approve.

When	Direction	Rate
Clinic hours, 08:30 – 21:00	Inbound	~15 Mbit/s sustained
Overnight	Outbound	~20 GB, about an hour at 50 Mbit/s
Each clinic site (2 rooms)	Outbound from the clinic	~1.4 Mbit/s

8 · Operational policy

Ask	Why
No automatic reboots or patch windows 08:00–21:30, six days a week	A reboot mid-consultation is survivable — the clinic PCs buffer and retry — but it costs recordings that cannot be re-taken
The archive volume exempt from quota sweeps and automatic cleanup	It grows continuously, by design
Disks encrypted at rest; archive readable only by the services and named staff	It is patient-identifiable health information
A UPS that shuts the machine down cleanly	A recording being written during an unclean power loss can be damaged
Somewhere for monitoring mail to go	Alerts on disk space, failed jobs, silent recorders and certificate expiry

9 · What we do not need

Worth stating, because it narrows the approval considerably: no inbound access to the database, ever · no public file shares, FTP or SMB · no access to any other UIU system, network or directory · no Windows licences · no CITS involvement in day-to-day running · no credentials held on the archive machine beyond one shared key, so a compromise there cannot enumerate storage or forge an integrity receipt.

10 · How we will prove it before a patient is recorded

Step	What it checks
1. Channel B test	CMED's two message types against the server's own rules
2. The 28-room storm	Twice the real concurrency, 150 MB per room; must lose no piece
3. Per-PC preflight	Each clinic machine, including listening to its microphone
4. A restore	Fetch one encrypted cloud copy and rebuild it. Day one, then monthly
5. One real consultation	End to end, before any patient is in the room

Steps 2 and 5 have already been done on the bench. A consultation has been carried from the microphone through enrolment, authorisation, segmenting, upload, verification, archiving, the clinical JSON, the encrypted cloud copy and the purge receipt that let the PC delete its own audio — and the recording appeared on the operations dashboard about ten seconds after Stop. What remains is hardware.

11 · The short version

We need	Detail
A machine	8 cores, 8 GB on Ubuntu Server 24.04 , 1 TB NVMe
Archive storage	1.5 TB usable minimum; 2 TB comfortable. RAID is for uptime, not safety — the off-site copy is the second copy
Three mount points	/srv/aims/archive · /srv/aims/db · /srv/aims/backups
A DNS name	lower case, static address
Inbound	443 TCP+UDP, 80 TCP for certificates, SSH from named machines
Outbound	443 to two storage providers, NTP, container registries
No middlebox	No TLS interception, no short idle timeouts
Policy	No patch window in clinic hours, no quota sweep on the array, UPS
A decision from UIU	How long audio is retained — it sets the storage bill
An answer from CITS	What disks the machine has , and who supports the Linux host

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